

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 12

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 The points $A(1, 2)$, $B(5, 4)$ and $C(k, 8)$ are collinear. Find k .

A 12

B 11

C 10

D 9

E 7

F 13

2 The polynomial $x^3 + ax^2 + bx + c$ has roots 2, -1 and 3. Find the value of $a + b + c$.

A 5

B -5

C -3

D 3

E -7

F 7

3 Let $f(x) = x^2 - 2x - 8$. For how many integers n is $f(n) < 0$?

A 8

B 6

C 5

D 7

E 3

F 4

- 4 A student attempts to solve the equation $\sqrt{x+3} = x-3$ for real x .
- I Squaring both sides: $x+3 = (x-3)^2 = x^2 - 6x + 9$.
 - II So $x^2 - 7x + 6 = 0$.
 - III So $(x-1)(x-6) = 0$.
 - IV Hence the solutions are $x = 1$ and $x = 6$.

Which of the following best describes the solution?

- A The solution is wrong, and the first error occurs on line IV.
- B The solution is completely correct.
- C The solution is wrong, and the first error occurs on line I.
- D The solution is wrong, and the first error occurs on line II.
- E The solution is wrong, and the first error occurs on line III.

- 5 The trapezium rule with n strips is used to estimate $\int_0^1 \sqrt{x} \, dx$. Which of the following is true for every n ?
- A The estimate is an overestimate.
 - B The estimate is an underestimate.
 - C The estimate is an underestimate only when n is even.
 - D The estimate is exact.
 - E It cannot be decided without calculating.

- 6 To prove **if n^2 is even then n is even** by contradiction, which assumption should be made?
- A n is odd.
 - B n^2 is even and n is odd.
 - C n^2 is even.
 - D n^2 is odd.
 - E n is even and n^2 is odd.

- 7 $ABCD$ is a quadrilateral. P : 'the diagonals of $ABCD$ bisect each other'. Q : ' $ABCD$ is a rectangle'. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .

- 8** Which of the following is the negation of the statement *******'every student in the class scored more than 50'*******?
- A** At least one student in the class scored more than 50.
 - B** Every student in the class scored 50 or less.
 - C** Every student in the class scored less than 50.
 - D** At least one student in the class scored 50 or less.
 - E** No student in the class scored more than 50.

- 9** $p(x)$ is a polynomial. P : ' $p(2) = 0$ '. Q : ' $(x - 2)^2$ is a factor of $p(x)$ '. Which one of the following is true?
- A** P is necessary and sufficient for Q .
 - B** P is necessary but not sufficient for Q .
 - C** P is sufficient but not necessary for Q .
 - D** P is neither necessary nor sufficient for Q .

10 Two circles have equations $x^2 + y^2 = 25$ and $(x - 8)^2 + y^2 = 9$. Which of the following statements is/are true?

- I** The circles touch externally.
- II** The circles intersect at exactly two points.
- III** The point $(5, 0)$ lies on both circles.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 11** How many positive integers less than 1000 are divisible by neither 2 nor 5?
- A** 399
 - B** 600
 - C** 500
 - D** 400
 - E** 450
 - F** 300

12 Which of the following is/are true for all real x ?

I $\sin^2 x + \cos^2 x = 1$

II $\sin(2x) = 2\sin x$

III $\sin(x + \pi) = -\sin x$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

13 Which of the following statements about $f(x) = x^3 - 3x$ is/are true?

I f is increasing for all $x > 1$.

II f has a local maximum at $x = -1$.

III $f(x) \geq -2$ for all real x .

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

14 x and y are positive integers with $x^2 - y^2 = 45$. How many possible pairs (x, y) are there?

A 0

B 6

C 2

D 1

E 3

F 4

- 15** A student notices that $1 + 2 = 3$, $4 + 5 + 6 = 7 + 8$, $9 + 10 + 11 + 12 = 13 + 14 + 15$ and conjectures that the pattern continues. Which of the following describes the n th line of the pattern?
- A** The n th line starts at n^2 and has $n + 1$ numbers on the left and n on the right.
- B** The n th line starts at $n(n + 1)$ and has $n + 1$ numbers on the left.
- C** The n th line starts at n^2 and has n numbers on the left and $n + 1$ on the right.
- D** The n th line starts at n^2 and has n numbers on each side.
- E** The n th line starts at $2n - 1$ and has $n + 1$ numbers on the left.

16 A selection of n distinct integers is taken from $1, 2, \dots, 20$. What is the smallest n that guarantees two of the chosen integers differ by exactly 10?

- A** 2
- B** 15
- C** 12
- D** 11
- E** 20
- F** 10

17 Claim: if $f'(a) = 0$ and $f''(a) = 0$ then f has a point of inflexion at $x = a$.

Which of the following shows the claim is false?

A $f(x) = x^3 - 3x$ at $a = 1$

B $f(x) = x$ at $a = 0$

C $f(x) = x^2$ at $a = 0$

D $f(x) = x^3$ at $a = 0$

E $f(x) = x^4$ at $a = 0$

18 Here is a proof that there are infinitely many primes, with the steps out of order.

I Then N leaves remainder 1 on division by each of $p_1, \dots, p_n$, so no p_i divides N .

II Suppose there are only finitely many primes $p_1, p_2, \dots, p_n$, and let $N = p_1 p_2 \cdots p_n + 1$.

III So N has a prime factor that is not in the list, which is a contradiction.

IV But $N > 1$, so N has at least one prime factor.

Which is the correct order?

A II, I, IV, III

B II, III, I, IV

C I, II, IV, III

D II, I, III, IV

E IV, II, I, III

19 Which of the following statements is/are true?

I $\binom{n}{r} = \binom{n}{n-r}$ for all integers $0 \leq r \leq n$

II $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$ for all integers $0 \leq r < n$

III $\sum_{r=0}^n \binom{n}{r} = 2^n$ for all positive integers n

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

20 Claim: the function $f(x) = x^3 - 3x^2 + 3x$ is increasing for all real x .

I $f'(x) = 3x^2 - 6x + 3 = 3(x - 1)^2$.

II So $f'(x) \geq 0$ for all x , with equality only at $x = 1$.

III Since $f'(x) > 0$ for all x , f is increasing.

Which of the following is true?

- A** Line III overstates what was shown: $f'(1) = 0$; but f is still increasing for all x , so the claim is true and the proof needs a better justification.
- B** The proof is completely correct.
- C** The first error is on line I.
- D** The first error is on line II: $f'(x)$ can be negative.
- E** The claim is false: f has a turning point at $x = 1$.

END OF TEST